
Assignment 2

02450 Machine Learning Spring 24

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GROUP 510

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Student ID	Regression part A	Regression part B	Classification	Exam Questions
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Table 1: Contributions of the three collaborators for this report.

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0 Introduction

In this report, wine data will be examined using various methods and models. In the first part of the report, one attribute from the wine dataset will be selected to be predicted based on the other attributes in the dataset. The data will then undergo some feature transformations to prepare it for determining the optimal regularization parameter, λ . Followed by an explanation which will focus on how the output attribute is computed based on the weights of the other attributes, determined by the lowest generalization error. In the second part, three models will be compared: regularized linear regression, artificial neural network (ANN), and a baseline. Using cross-validation, it will be determined which of the models performs better and assess whether they outperform the baseline. To determine the models a table will be created showing the test error (E_i^{test}), hidden units (h_i^*) and regularization strength (λ_i^*) will be made. As well as the table the three models will also be evaluated statistically to determine if one of the models outperformed the other two. Lastly a multi classification problem based on the wine data will be solved. Here logistic regression, multi class and a baseline will be compared, like part 2 the output of the comparison will be a table with test error (E_i^{test}), hidden units (x_i^*) and regularization strength (λ_i^*). Lastly a logistic regression model will train on the wine data and using an optimal λ , explain the model makes a prediction.

1 Regression

1.1 Focus of regression

The goal of the linear regression is to predict the color intensity of the wine data based on the other 12 attributes seen in table 2 (The Cultivars have been excluded). The Color intensity was chosen to be predicted, because it could be interesting if we could predict if it was a white wine or red wine. Before performing linear regression, a feature transformation has been employed, namely, the data has been standardized. The data are been standardized during each fold of the cross-validation.

1.2 Regularization

The point of introducing the regularization parameter are to reduce overfitting and reduce variance of the estimated parameters. In general when λ is large models have low variance and tend to have large bias, when λ is small model tend to have high variance and small bias, so to find the best model an optimal λ would be required. By test for a range of λ s, an optimal λ have been found for the model. The range for which λ are chosen to be tested are from 1 to 100000, where λ scale logarithmic. The test error for different values of λ have been estimate and can be seen in table 3 A plot of the estimated generalization error have been made and can be seen in figure 1. From figure 1, it is observed that the validation error first drops slightly, then increases as λ increases. It is therefore observed that the optimal λ is indeed 10, as it is where the errors are at their lowest.

Attribute	Description
1) Cultivars	Which vineyard the wine comes from
2) Alcohol	Percentage of alcohol in wine.
3) Malic acid	Grams of malic acid per liter.
4) Ash	Ash content in grams per liter.
5) Alkalinity of ash	Alkalinity of ash in mEq/l.
6) Magnesium	Magnesium content in mg/dm ³ .
7) Total phenols	Total phenolic content in mg/dm ³ .
8) Flavanoids	Flavanoid content in mg/dm ³ .
9) Nonflavanoid phenols	Nonflavanoid phenolic content in mg/dm ³ .
10) Proanthocyanins	Proanthocyanin content in mg/dm ³ .
11) Color intensity	Color intensity.
12) Hue	Hue.
13) OD280/OD315 of diluted wines	OD280/OD315 ratio of diluted wines.
14) Proline	Proline content in mg/dm ³ .

Table 2: Attributes of Wine

λ	Test Error
1	0.416
10	0.408
100	0.458
1000	0.615
10000	0.714
100000	0.731

Table 3: Error values for different lambda

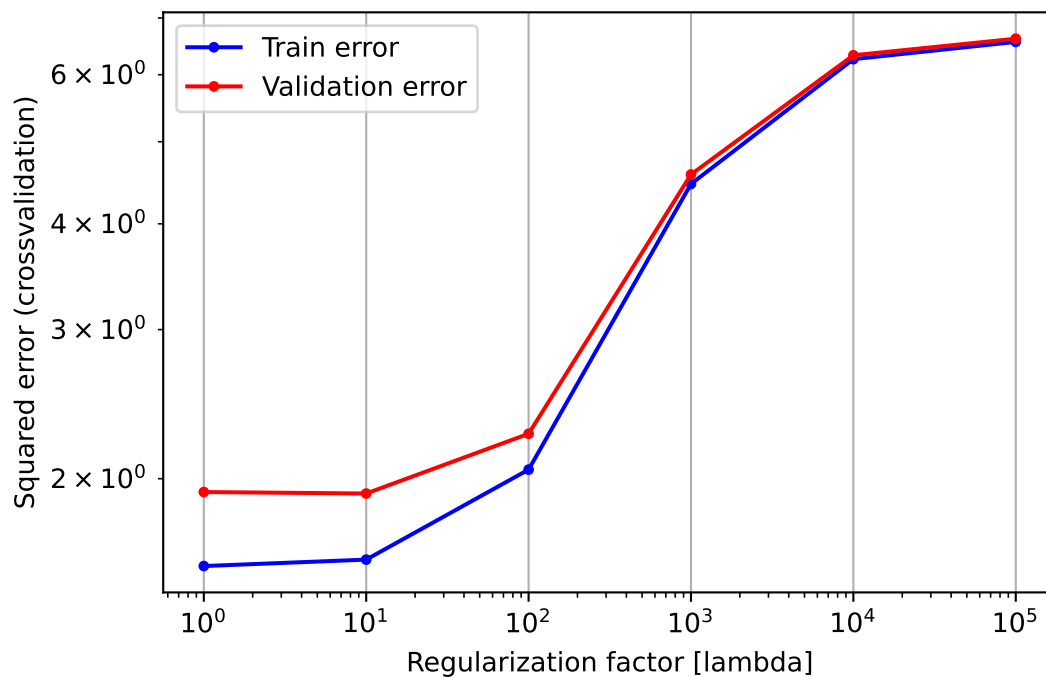


Figure 1: Estimated generalization error

1.3 Cross-validation

From table 4 we can see how the different attributes effect the Color intensity. We see some of the values are negative, when attributes have values $0 >$ it means that the attribute positively affect the Color intensity when you decrease them. From the table it is seen that Cultivars, Alcohol, Flavanoids and Proline are the attributes with the highest weights predicting the Color intensity. From Assignment 1 we did a heat map of the correlation between all the attributes and here we see the 2 most correlated with Color intensity are Alcohol and Proline. However Flavanoids are negative correlated. We can also see that the Hue have a negative weight and from Assignment 1 we see that it is negative correlated with the Color intensity. Therefor overall the weights of the attributes makes sense based on our findings in the first report.

Cultivars	2.55
Alcohol	0.91
Malic acid	-0.15
Ash	0.29
Alcalinity of ash	-0.41
Magnesium	-0.02
Total phenols	0.09
Flavanoids	0.7
Nonflavanoid phenols	0.07
Proanthocyanins	0.17
Hue	-0.55
OD280/OD315diluted	-0.32
Proline	0.8

Table 4: Weights in the last fold

1.4 Two-level Cross-validation and Comparison of models

This part's objective is to compare three different models' performance in predicting the wine's color intensity. We will use a baseline (Dummy), a regularized linear regression and an artificial neural network model (ANN). The Dummy model simply predicts the output by computing the average of the i 'th training dataset given in the outer fold i . In order to do so we implement a two-level cross-validation. As complexity controlling parameters for the ANN and the regularized models, we use respectively the number of hidden units and the regularization parameter λ .

We first wanted to know what range of values for the complexity controlling parameters was interesting to use in the implementation. So we implemented a simple K-fold cross-validation and manually changed the values for h and λ in the ANN and Regularized Model. After a few runs we selected the following values for λ : [0.01, 0.1, 1, 10, 100] and for h : [1, 2, 3, 4, 5].

Now that we have the values we want to use, we implement our two-level cross-validation with a number of $K1 = 10$ outer folds and $K2 = 5$ inner folds. In the inner fold, we train 5 ANN and 5 Regularized models on the j 'th training inner dataset given in the inner fold j . In each j we also go through the 5 ANN and Regularized models by going trying the different values of h and λ on the inner dataset. This allows us to confidently select the optimal h and λ for the current outer fold. We then train our three models on the training dataset from the outer fold using the parameters found with the inner fold. Table 5 displays the results we found from our implementation, which includes the generalization error E_i^{test} from each model using the optimal parameters for each outer fold i .

Outer fold	ANN		Linear Regression		Baseline
	Hidden layers n_i^*	E_i^{test}	λ_i^*	E_i^{test}	E_i^{test}
1	2	0.96	1	1.23	3.64
2	2	1.96	10	3.75	4.80
3	1	2.84	0.01	2.99	7.37
4	2	2.92	1	1.79	6.09
5	2	1.95	10	1.88	2.91
6	2	1.64	10	2.20	5.18
7	1	2.41	10	3.05	6.08
8	2	2.99	10	1.31	2.69
9	2	3.14	10	2.10	5.66
10	1	2.83	100	2.83	9.62

Table 5: Table containing the optimal models and corresponding estimated generalization error from each outer fold.

As found in the previous section, for most of the outer training datasets the optimal regularization parameter for the linear regression λ is equal to 10. We can also easily see that both the ANN and Regularized models are better at predicting the color intensity values compared to the Baseline as they have a lower generalization error. Nevertheless we expected this outcome as our models should always be more efficient then the Baseline. The more interesting comparison is between the ANN and Regularized models. It is however hard to tell just by looking at the table which performs better, we then do a statistical evaluation of our results.

1.5 Statistical evaluation

We want to evaluate if there's a significant performance difference between the fitted ANN, linear regression model and baseline. For that, we chose the test **setupI** for the statistical evaluation. Table 6 displays the results of the pairwise comparisons between the models (Baseline vs Linear Regression, Baseline vs Artificial Neural Network and Linear Regression vs Artificial Neural Network) computing the confidence intervals and the p-values.

Model comparing	p-value	Confidence interval	
		$CI-$	$CI+$
Baseline / Linear Regression	$6.64 \cdot 10^{-5}$	1.84	3.62
Baseline / ANN	$7.11 \cdot 10^{-4}$	1.67	4.40
Linear Regression / ANN	0.54	-0.79	1.41

Table 6: Statistical Evaluation of our three models comparing them pairwise by computing the resulting p-values and confidence intervals with ($\alpha=0.05$)

Let's breakdown the three comparisons one by one:

Baseline vs Linear Regression:

The p-value here is very small (approximately $6.64e-05$). This indicates strong evidence against the null hypothesis, suggesting that the mean squared error of the Linear Regression is significantly different from that of the Baseline model.

The CI for the difference between the Baseline and Linear Regression ranges from approximately 1.84 to 3.62. This means that we are 95% confident that the true difference in mean squared error between the Baseline and Linear Regression falls within this interval. As the values are positive, we can safely say that the Linear Regression model performs better than the Baseline as we expected.

Baseline vs Artificial Neural Network:

The p-value for comparing the Baseline model to the ANN model is also small (approximately 0.0007), indicating strong evidence against the null hypothesis. This suggests that the mean squared error of the ANN model is significantly different from that of the Baseline model.

The CI for the difference between the Baseline and ANN models ranges from approximately 1.67 to 4.41. This means that we are 95% confident that the true difference in mean squared error between the Baseline and ANN models falls within this interval. Same as before, we can safely say that the Artificial Neural Network performs better than the Baseline as we also expected.

Linear Regression vs Artificial Neural Network:

The p-value for comparing the Regularized model to the ANN model is relatively large (approximately 0.543), indicating weak evidence against the null hypothesis. This suggests that there is no significant difference in the mean squared error between the Linear Regression and ANN models.

The CI for the difference between the Regularized and ANN models ranges from approximately -0.79 to 1.41. This interval contains 0 indicating that there's a possibility the two models are identical when it comes to the performance.

Based on what we've learned, here what we could do for further studies: Improve the performance of the Artificial Neural Network by fine-tuning the architecture of our model. Indeed as for now, the Linear Regression model and Artificial Neural Network model performances are quite similar, making it hard to choose which is best for regression on our dataset and predict the color intensity of the wine. For better predictions and performance, we could use ensemble methods such as bagging, boosting, or stacking in order to combine the predictions of multiple models, and so there strengths and minimizing there weaknesses. In Table 5, we indeed see for different outer folds that it's the optimal model changes between the Artificial Neural Network and the Linear Regression.

2 Classification

2.1 Focus of Classification

The classification task is chosen to be a prediction of which Cultivar a wine belongs to. The multi-class classification problem is trained on all the attributes in the wine dataset and performed with baseline, logistic regression and an ANN.

2.2 Comparing the 3 models (1/2)

Three classification models, namely multinomial regression, ANN, and a baseline are compared as previously done in subsection 1.4. Again, a 2-level cross fold is used for splitting the dataset into a training data part and a test data part. The training data is by the inner fold divided further into a training dataset and a validation test dataset. In these inner folds, the models are fit to the training data and the validation error calculated to be compared against each other. The model with the lowest validation error is chosen and trained on the outer fold training data and tested on the test data.

The ANN is investigated with the number of hidden layers going from 1 to 5. The multinomial regression has 5 different regularization strengths going from 0.01 to 100 by factor 10. The baseline is used on the outer folds, as it will predict the test data to have one class, which was the one most occurring in its training data.

2.3 Comparing the 3 models (2/2)

The script was run with the described ranges of the complexity-controlling parameters, and the result in classification is calculated upon if an observation is classified correctly. Therefore, the error is calculated as:

$$E = \frac{\text{Number of misclassified obs}}{N_{test}} \quad (1)$$

Each outer fold has an optimal model for each type of model and the corresponding estimated generalization error. These are depicted in Table 7.

2.4 Statistical evaluation

The three models are compared pairwise utilizing setup 2 (the correlated t-test), where the estimated generalization errors from all the outer splits are used as the values of the loss-function. Therefore, with 10 outer folds, 10 different r's are used for the the function `correlated_ttest` which outputs p-values and confidence intervals comparing the different models. The result is shown here in Table 8

Outer fold	ANN		Multinomial Regression		Baseline
	Hidden layers n_i^*	E_i^{test}	λ_i^*	E_i^{test}	E_i^{test}
1	2	0.0%	0.1	5.6%	61.1%
2	4	5.6%	100.0	27.8%	66.7%
3	2	11.1%	0.1	11.1%	61.1%
4	4	0.0%	10.0	0.0%	44.4%
5	3	5.6%	1.0	5.6%	50.0%
6	3	0.0%	0.0	0.0%	55.6%
7	2	5.6%	0.0	5.6%	55.6%
8	3	0.0%	100.0	16.7%	66.7%
9	4	11.8%	0.1	11.8%	64.7%
10	2	5.9%	0.0	0.0%	76.5%

Table 7: Table containing the optimal models and corresponding estimated generalization error from each outer fold.

Model comparing	p-value	Confidence interval	
		$-\theta$	$+\theta$
Baseline / Multinomial Regression	$0.093 \cdot 10^{-9}$	41.8%	61.8%
Baseline / ANN	$0.018 \cdot 10^{-9}$	46.8%	64.5%
Multinomial Regression / ANN	0.34	-4.77%	12.5%

Table 8: Table containing the optimal models and corresponding estimated generalization error from each outer fold.

It is clear that the baseline is the worst model, and both Multinomial Regression and ANN is better, since the p-value is lower than the confidence level of 0.05. The ANN and Multinomial regression are quite close, and the null-hypothesis that the two models are equally good can't be turned down with a p-value over 0.05 which it is. The confidence intervals for the multinomial and ANN show that there is a weighting towards that the ANN is better, since the positive confidence level is higher, however this cannot be proved on this data.

We have learned that it is important to give a number of replicates for the training of neural network, since the initialization means a lot for finding the optimum. With more initializations, more folds and maybe a larger dataset, even better classifications could be done, however, our script took 1.5 hours to run, so we didn't push the calculations further. We learned that compared to the baseline, it is actually achievable doing classification on a series of different attributes.

3 Discussion

3.1 What we have learned

From the regression, we have both learned something about different models for regression and for our case, prediction of the color intensity of the wine. It is possible to train the models on the data and predict the color intensity of the wine with both an artificial neural network and a regularized linear regression. Which model is the best can't be determine from this and perhaps our data is not complex enough so that the neural network could show it's ability to handle very complex data. We also learned about the regression parameter λ and the number of hidden units h and how the magnitude of those parameters can influence how well the model fits to the data.

The classification task taught us about multi-class classification especially, since there are 3 cultivars to predict for a origin of a given wine. It was shown that it was possible to determine the origin with some uncertainty. Some of the trained models performed with no mis-classifications, others had some. The training sets are quite short with 178 wines being divided into 10 outer fold and 10 inner folds, so more training data might give the models better performance overall.

3.2 Previous study analysis

The wine data has previously been used as validation for regularized discriminant analysis as a classifier. For RDA in [1], they found that their model performed better for a regularization parameter which were negative. Where the optimal regularization parameter found in this report was positive.

- Previously the data has been used for comparing different classifiers in [2]. If one reads the article, it compares 8 different methods e.g. LDA (Linear Discriminant Analysis), RDA (Regularized Discriminant Analysis), and 1NN(k-Nearest Neighbour with k equal to 1). It is concluded that only RDA achieved 100% on classifying data from the Wine Dataset used in this group.
- In the article [1] *The data was used as an illustrator of superior performance for a new appreciation function using Regularized Discriminant Analysis. The dataset was also used to compare Error counting (EC) and The geometric appreciation function (GAF).*

4 Problems

4.1 Problem 1

From the ROC curve, what subplot A,B,C or D corresponds to the ROC curve. Looking at the curve there are 5 steps where the curve changes. From the 5 steps we can get the right subplot From step 1 $TPR = 0.25$ and $FPR = 0$

Step 2 the change are $\text{TPR} = 0.25$ and $\text{FPR} = 0.5$

Step 3 the changes are $\text{TPR} = 0.75$ and $\text{FPR} = 0.5$

Step 4 the changes are $\text{TPR} = 0.75$ and $\text{FPR} = 1$

Step 5 we get $\text{TPR} = \text{FPR} = 1$

Changing the steps to an array we get

$$[, 0, 0, 1, 1, 0, 0, 1]$$

From the array we see it is the same as C, so the answer are C

4.2 Problem 2

To calculate the impurity equations (9.1) and (9.4) from the book have been used. Before the split there are $N = 135$ observations, after the first split there are $N_1 = 120$ observations in split $x_7 = 0$ and $N_2 = 15$ observations in split $x_7 = 1, 2$. The different impurity functions are calculated as follows

$$I_0 = 1 - \frac{37}{135}$$

$$I_1 = 1 - \frac{33}{120}$$

$$I_2 = 1 - \frac{5}{15}$$

And the ClassError impurity measure are calculated as below

$$\Delta = I_0 - \frac{N_1}{N}I_1 - \frac{N_2}{N}I_2$$

Which gives the impurity for split 2 as $\Delta = 0.0074$, so the answer are C

4.3 Problem 3

Determine the number of parameters in a artificial neural network trained on a Urban Traffic dataset. From the question description we find that there are $M = 7$ attributes of the dataset. There are one hidden layer with $n_h = 10$ units and there are $C = 4$ multi class outputs.

$$(M + 1)n_h + (n_h + 1)C = 124$$

The Network contains 124 parameters, so the answer to the question are A

4.4 Problem 4

For question 4 we need to determine the correct rule for assigning nodes in a decision tree. To determine the correct rule, one start at the top of the decision tree and see if the decisions are a correctly. When checking the different answers, it is seen that the one which assign the nodes correctly are the answer A

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